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Comparative Analysis of Lipid Profiles Between Controlled and Uncontrolled Type 2 Diabetes Mellitus Patients at Tarakan Regional General Hospital (2022–2023)

ABSTRACT

Background: Type 2 diabetes mellitus (T2DM) contributes substantially to cardiovascular complications, largely through disturbances in lipid metabolism associated with inadequate glycemic control. However, evidence from hospital-based populations in Indonesia remains limited. This study aimed to evaluate differences in lipid profiles between patients with controlled and uncontrolled T2DM.

Methods: A cross-sectional analytic study was conducted using medical records of 120 patients with T2DM. Patients were classified into controlled (HbA1c <7%) and uncontrolled (HbA1c ≥7%) groups. Lipid parameters assessed included total cholesterol, LDL-C, HDL-C, and triglycerides. Statistical analysis was performed using appropriate parametric and non-parametric tests.

Results: Patients with uncontrolled T2DM exhibited significantly higher total cholesterol, LDL-C, and triglyceride levels compared with controlled patients. No statistically significant difference was observed in HDL-C levels. Dyslipidemia was more prevalent in patients with poor glycemic control.

Conclusion: Uncontrolled T2DM is associated with unfavorable lipid alterations, emphasizing the importance of maintaining optimal glycemic control to reduce cardiovascular risk.

Keywords: diabetes mellitus, HDL, total cholesterol, LDL, triglycerides

INTRODUCTION

Type 2 diabetes mellitus (T2DM) has become an increasingly important global health issue due to its strong association with cardiovascular morbidity and mortality. According to the International Diabetes Federation, Indonesia is among the countries with a high number of individuals affected by diabetes, with an estimated 19.5 million cases reported in 2021.¹ National data further indicate a continuous rise in prevalence, increasing from 6.9% in 2013 to 8.5% in 2018.² This growing trend reflects a substantial burden on public health systems.

The underlying mechanisms of T2DM involve insulin resistance and persistent hyperglycemia, both of which disrupt normal lipid metabolism. These disturbances often lead to diabetic dyslipidemia, typically characterized by elevated triglycerides, increased low-density lipoprotein cholesterol (LDL-C), and decreased high-density lipoprotein cholesterol (HDL-C).³ Such alterations contribute to the development of atherosclerosis and increase the risk of cardiovascular complications.³⁻⁵ In addition, prolonged hyperglycemia may induce oxidative stress and endothelial dysfunction, further aggravating vascular damage.³⁻⁵

Glycemic control, commonly evaluated using glycated hemoglobin (HbA1c), plays a crucial role in determining metabolic outcomes in patients with T2DM. Individuals with inadequate glycemic control (HbA1c $\geq 7\%$) are more likely to exhibit unfavorable lipid profiles compared to those with controlled glycemic status (HbA1c $< 7\%$).^{4,5} Despite the well-recognized association between glycemic status and lipid abnormalities, evidence from hospital-based populations in Indonesia remains limited.

Therefore, this study was conducted to compare lipid profiles between patients with controlled and uncontrolled T2DM and to assess their clinical implications, particularly in relation to cardiovascular risk.

METHODS

This analytic cross-sectional study was conducted at a tertiary care hospital in Jakarta, Indonesia, in October 2024. Ethical approval was obtained from the Institutional Ethics Committee (No. 1811/SLKE/TM/UKK W/FKIK/KEPK/TX/2024).

A total of 120 patients with type 2 diabetes mellitus (T2DM) were included using consecutive sampling from medical records. T2DM was defined based on documented glycated hemoglobin (HbA1c) $\geq 6.5\%$ or fasting plasma glucose ≥ 126 mg/dL. Patients with incomplete lipid profile data, those receiving lipid-lowering therapy (including statins, fibrates, niacin, or ezetimibe), and those with chronic kidney disease, hyperthyroidism, or liver cirrhosis were excluded.

Patients were categorized into controlled (HbA1c $< 7\%$) and uncontrolled (HbA1c $\geq 7\%$) groups. The independent variable was glycemic control status, while the dependent variables included lipid parameters: triglycerides, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and total cholesterol. Lipid profile measurements were obtained from medical records and were assessed within a clinically relevant timeframe relative to glycemic evaluation.

Data were analyzed using SPSS version 24. Normality of data distribution was assessed prior to analysis. Independent t-test was used for normally distributed variables, while the Mann-Whitney U test was applied for non-normally distributed data. A p-value < 0.05 was considered statistically significant.

RESULTS

A total of 120 patients with type 2 diabetes mellitus (T2DM) were included in this study. The variables presented in Table 1 include demographic characteristics (age and sex) and anthropometric measurements, specifically body mass index (BMI). Most participants were in the pre-elderly age group (45–59 years), comprising 62 individuals (52.7%). Females represented the majority of the cohort (n = 73; 60.8%). Based on body mass index (BMI), Obesity Class I (25–29.9 kg/m²) was the most prevalent category (n = 49; 40.8%). Furthermore, uncontrolled diabetes was observed in 77 patients (64.2%), indicating that poor glycemic control predominated in this population.

Table 1. Distribution of Study Participants According to Demographic and Clinical Characteristics

Characteristics	Frequency (%)	Cholesterol Level			
		Total Cholesterol	Triglycerides	LDL	HDL
		(mg/dL)			
Age(years)					
Adult(19-44)	10 (8,3)	182,5 (105-324)	130 (68-193)	119 (60-225)	42 (12-60)
Pre-Elderly(45-59)	62 (52,7)	212 (108-323)	160 (57-828)	121 (46-211)	46 (23-75)
Elderly (≥60)	48 (40,0)	191 (95-299)	131 (49-366)	106 (19-209)	45 (23-79)
Sex					
Male	47 (39,2)	177 (95-315)	142 (49-828)	110 (19-209)	44 (12-73)
Female	73 (60,8)	216 (123-343)	143 (57-617)	129 (45-225)	50 (23-79)
BMI (kg/m²)					
Underweight (<18,5)	7 (5,8)	205 (145-218)	195 (75-285)	121 (80-144)	43 (26-60)
Normal (18,5-22,9)	14 (11,7)	182 (135-235)	116 (57-213)	101 (45-159)	43 (27-79)
Overweight(23-24,9)	24 (20,0)	186 (108-343)	129 (49-480)	108 (46-209)	45 (23-75)
Obesity Class I (25-29,9)	49 (40,8)	212 (95-324)	149 (84-828)	124 (19-225)	47 (12-75)
Obesity Class II (≥30)	26 (21,7)	194 (123-264)	150 (66-385)	117 (75-211)	47 (23-74)
Glycemic Control Status					
Controlled (< 7%)	43 (35,8)	179,23 ±43,24	128 (49-385)	109,70 ±38,66	45,86 ±13,66
Uncontrolled (≥ 7%)	77 (64,2)	210,86 ±51,64	156 (57-828)	122,62 ±37,72	47,29 ±13,68

Analysis of lipid parameters across age groups showed that the pre-elderly group exhibited the highest median concentrations of total cholesterol [212 (108–323) mg/dL], triglycerides [160 (57–828) mg/dL], LDL-C [121 (46–211) mg/dL], and HDL-C [46 (23–75) mg/dL], suggesting a clustering of lipid abnormalities within this age category. When stratified by sex, female patients demonstrated higher median lipid levels compared with males. Median total cholesterol reached 216 (123–343) mg/dL and LDL-C 129 (45–225) mg/dL among females, indicating a less favorable lipid profile in this subgroup.

Across BMI categories, patients classified as Obesity Class I had the highest median total cholesterol and LDL-C levels, measuring 212 (95–324) mg/dL and 124 (19–225) mg/dL, respectively. In contrast, the highest median triglyceride concentration was observed in the underweight group at 195 (75–285) mg/dL, demonstrating variability in lipid distribution across nutritional status. Stratification according to glycemic control revealed that patients with uncontrolled diabetes consistently exhibited higher lipid levels, including total cholesterol of 210 ± 51.64 mg/dL, triglycerides of 156 (57–838) mg/dL, and LDL-C of 122.62 ± 37.72 mg/dL.

Table 2. Distribution of Dyslipidemia Among Patients with Type 2 Diabetes Mellitus

Dyslipidemia	Glycemic Control Status	
	Controlled (n=43)	Uncontrolled (n=77)
	n (%)	
Dyslipidemia	4 (3,3)	12 (10,0)

As shown in Table 2, dyslipidemia was more prevalent among patients with uncontrolled diabetes, accounting for 12 individuals (10.0%) of the study population.

Table 3. Comparison of Lipid Parameters According to Glycemic Control Status

Lipid Profile	Glycemic Control Status		Nilai p
	Controlled (n=43)	Uncontrolled (n=77)	
LDL	109,70 $\pm$ 38,66	122,62 $\pm$ 37,72	0,047 ^a
HDL	45,86 $\pm$ 13,66	47,29 $\pm$ 13,68	0,585 ^a
Kolesterol Total	179,23 $\pm$ 43,24	210,86 $\pm$ 51,64	0,001 ^a
Trigliserida	131,66 (49-285)	156,00 (57-828)	0,013 ^b

^aUji T test independent, ^bUji mann whitney, significant if $p < 0,05$

The results demonstrated significant differences in total cholesterol ($p = 0.001$), LDL-C ($p = 0.047$), and triglyceride levels ($p = 0.013$) between patients with controlled and uncontrolled type 2 diabetes mellitus. Overall, patients with uncontrolled diabetes exhibited higher lipid levels compared with those with controlled diabetes. The mean LDL-C level was 122.62 ± 37.72 mg/dL, and the mean total cholesterol level was 210.86 ± 51.64 mg/dL. Meanwhile, the median triglyceride level was 156.00 (57–828) mg/dL.

DISCUSSION

The findings of this study indicate that most patients with type 2 diabetes mellitus (T2DM) were within the pre-elderly age group (45–59 years), which is consistent with previous reports showing that metabolic disorders are more prevalent among individuals aged above 45 years.^{9,10} Age-related physiological changes, including reduced metabolic efficiency and lifestyle factors, may contribute to impaired regulation of glucose and lipid metabolism.^{10,11}

In the present study, female patients constituted the majority of the study population and tended to have higher lipid levels compared to males. This observation is consistent with previous findings suggesting that women with T2DM may have a higher tendency toward dyslipidemia.^{7,12} Hormonal influences, particularly reduced estrogen levels, may contribute to increased visceral fat accumulation and insulin resistance, thereby affecting lipid metabolism.¹¹

Regarding nutritional status, most patients were categorized²⁵ Obesity Class I, which was associated with elevated total cholesterol and LDL-C levels. These findings are in agreement with earlier studies that have demonstrated a strong association between obesity²³ and lipid abnormalities in T2DM patients.^{13, 14} Excess adipose tissue can increase the release of free fatty acids and pro-inflammatory cytokines, which may disrupt insulin signaling and enhance hepatic lipoprotein production, thereby worsening dyslipidemia.^{13, 15}

Interestingly, lipid abnormalities were also observed among underweight individuals, particularly in the form of elevated triglyceride levels. Although evidence remains limited, this suggests that dyslipidemia in T2DM is not exclusively related to obesity²³ but may also be influenced by metabolic stress and other underlying conditions.^{14, 16} This finding highlights the complex nature of lipid metabolism disturbances in diabetic patients.

The main finding of this study shows that patients with uncontrolled T2DM exhibited significantly higher levels of total cholesterol, LDL-C, and triglycerides compared with those with controlled glycemic status.¹³ These results are consistent with previous studies indicating that poor glycemic control is associated with a more atherogenic lipid profile.^{4, 5, 15} Increased LDL-C levels in uncontrolled diabetes may be linked to enhanced hepatic production of very low-density lipoprotein (VLDL) and the subsequent formation of atherogenic lipoprotein particles.^{18, 19}

In contrast, HDL-C levels did not differ significantly between the two groups, although slightly higher levels were observed in patients with uncontrolled diabetes. Previous studies generally report reduced HDL-C levels in T2DM due to increased catabolism of HDL particles and impaired reverse cholesterol transport.¹⁹ The absence of a significant difference in this study may be influenced by variations in patient characteristics, treatment adherence, or sample size.

Higher triglyceride levels observed in patients with poor glycemic control further support the role of insulin resistance in lipid metabolism disturbances. Insulin resistance promotes increased lipolysis and the release of free fatty acids, leading to enhanced hepatic triglyceride synthesis and contributing to dyslipidemia.¹⁹

Overall, the findings of this study emphasize the importance of maintaining optimal glycemic control in patients with T2DM. Individuals with uncontrolled diabetes tend to exhibit more atherogenic lipid profiles, characterized by elevated LDL-C, total cholesterol, and triglycerides. These findings support the need for integrated management strategies that include both glycemic control and routine lipid monitoring to reduce cardiovascular risk.²⁰

CONCLUSION

This study demonstrates that glycemic status plays a critical role in shaping lipid abnormalities in patients with type 2 diabetes mellitus. Uncontrolled diabetes was consistently associated with significantly higher levels of total cholesterol, LDL-C, and triglycerides, reflecting a more atherogenic lipid profile and a greater potential for cardiovascular complications. Notably, lipid disturbances were observed across different BMI categories, including underweight individuals, indicating that dyslipidemia in T2DM is not solely attributable to obesity but arises from complex metabolic interactions involving insulin resistance, glycemic control, and lipid metabolism. These findings underscore the importance of comprehensive metabolic assessment and strict glycemic management, alongside routine lipid monitoring, to reduce cardiometabolic risk and improve long-term outcomes in patients with T2DM.

7 **Conflict of Interest**

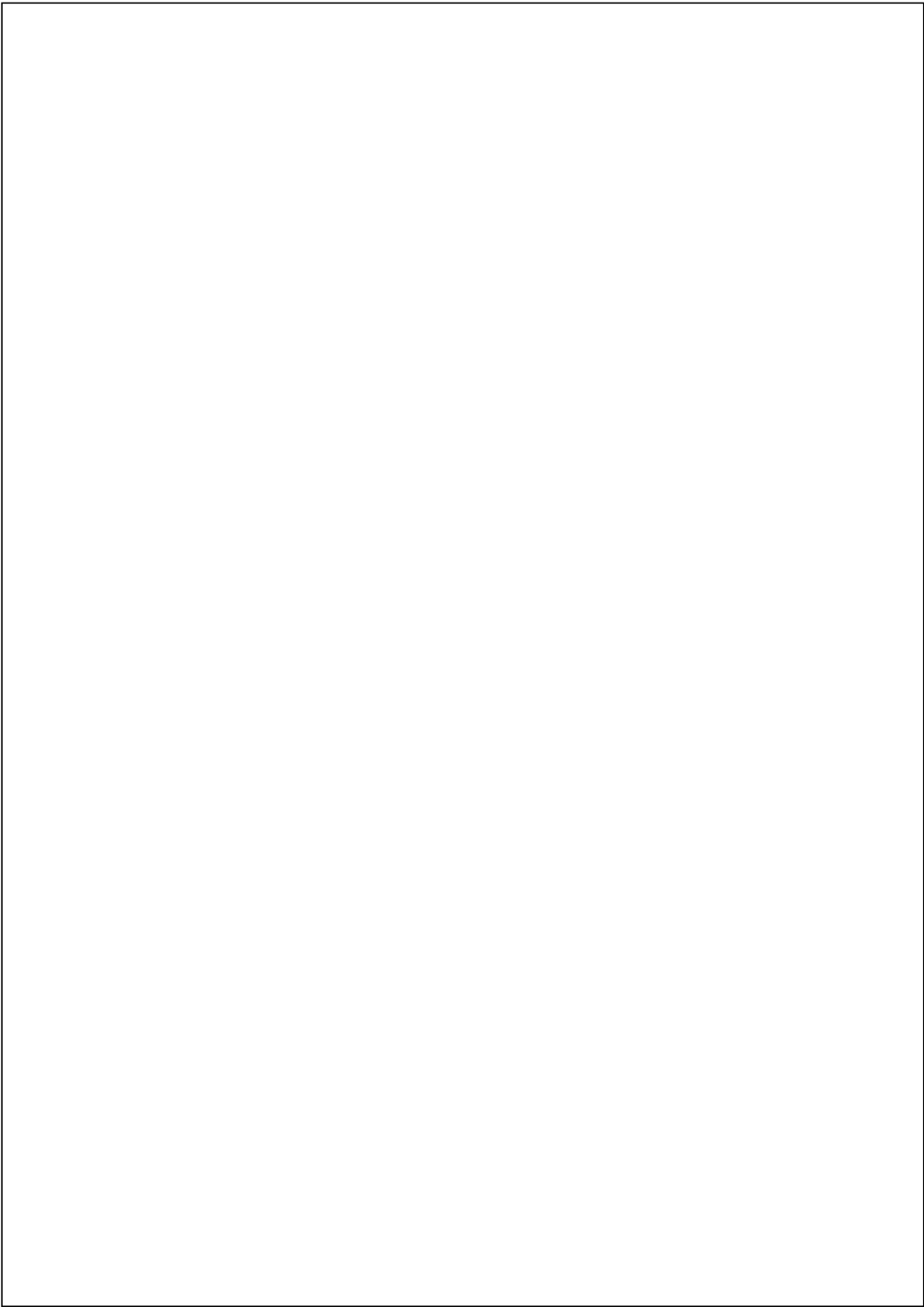
The authors declare that there are no conflicts of interest regarding the publication of this study.

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